

adb-auto-enable Usage Guide

Introduction

adb-auto-enable is an Android utility designed to automatically enable wireless ADB (Android Debug Bridge) debugging and switch it to a fixed port (**5555**) on every device boot.

Purpose

Android 14 and newer versions introduce enhanced security measures that randomize the ADB port and disable wireless debugging after a reboot or sleep. This behavior breaks automation setups for non-interactive devices such as Android TV boxes or Chromecasts.

adb-auto-enable solves this by providing a completely autonomous, **no-root-required** way to maintain persistent remote ADB access.

Prerequisites

- **Device:** Android 13 or higher (specifically tested on Chromecast with Google TV).
- **Network:** An active WiFi connection.
- **Security:** Use only on **isolated or trusted networks** (e.g., home network behind a firewall) as port 5555 provides full system privileges to authorized computers.

Installation

Method 1: Pre-built APK

1. Download the latest APK from the GitHub Releases page.
2. Sideload the APK onto your Android device.

Method 2: Build from Source

If you prefer to build it yourself, ensure you have the Android SDK installed:

```
git clone https://github.com/mouldybread/adb-auto-enable.git
cd adb-auto-enable
./gradlew assembleRelease
adb install app/build/outputs/apk/release/app-release.apk
```

Usage

The setup involves a one-time pairing process that does not require a PC.

1. Initial Pairing

1. Open the app on your device or navigate to `http://<device-ip>:9093` in a web browser from a device on the same network.
2. On the Android device, go to **Settings** → **Developer Options** → **Wireless Debugging**.
3. Select **“Pair device with pairing code”**.
4. Enter the provided **pairing code** and **port** into the web interface and click **“Pair Device”**.

2. Verification

1. The web UI should show the permission status as **“[OK] Granted”**.
2. Click **“Switch to Port 5555 Now”** to test.
3. The status for “ADB Port 5555” should change to **“[OK] Available”**.

3. Auto-Boot Test

1. Reboot the device.
2. Wait approximately 90 seconds (60s boot delay + 30s configuration time).
3. Connect from your computer using:

```
adb connect <device-ip>:5555
```

Troubleshooting

- Ensure the device is connected to the same WiFi network as the pairing computer.
- If the service doesn't start, check if the app has been granted the necessary permissions.
- Wireless debugging must be enabled in Developer Options for the initial pairing.

Source: <https://github.com/mouldybread/adb-auto-enable>